

Minor-2 Syllabus

1. Singly Linked Lists: Insertion, deletion, and concatenation operations
2. Circular Linked Lists: Basic operations
3. Doubly Linked Lists: Insertion and deletion operations
4. Stack ADT: Definition and operations, Array and linked list implementations in C
5. Stack Applications: infix to postfix conversion, postfix expression evaluation, recursion implementation
6. Queue ADT: Definition and operations, Array and linked list implementations in C

Hashing: The fastest search is the one you never perform.



Real World Searching Problem [2 billion users]

Instagram

Phone number, username, or email

Password

☐ Save login info

Log in

Forgot password?

Linear Search

0 idk_why_im_here
1 chai_over_wifi
2 maggi_in_2_hours
3 sleeping_is_my_cardio
4 error_404_me_not_found
5 born_to_nap
6 too_lazy_to_think
7 loading_username
8 crying_in_the_club
9 dont_follow_me_pls
10 sarcasm_loading

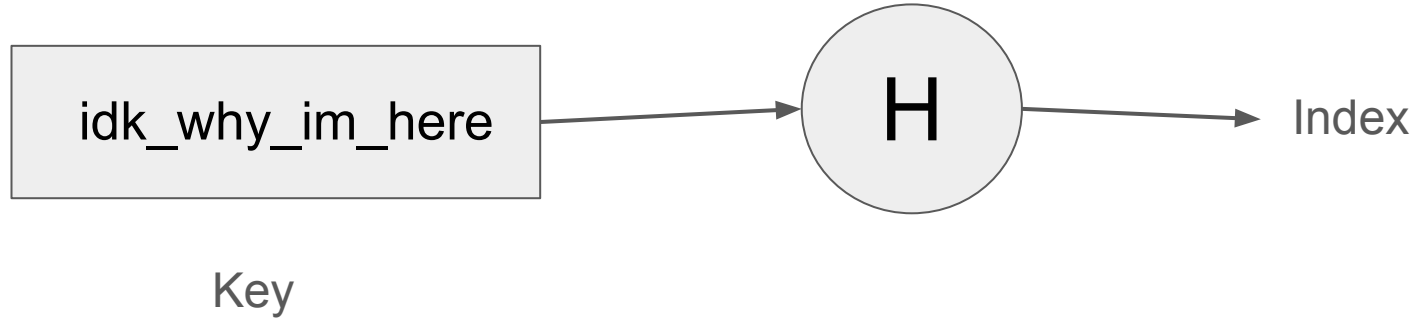
11 mood_swings_247
12 hash_me_if_you_can
13 bug_fixer_in_progress
14 ctrl_c_ctrl_sleep
15 caffeine_and_code
16 not_a_robot_trust_me
17 buffering_life
18 procrastination_pro
19 low_battery_human
20 brain_not_found

Binary Search

0	born_to_nap
1	brain_not_found
2	buffering_life
3	bug_fixer_in_progress
4	caffeine_and_code
5	chai_over_wifi
6	crying_in_the_club
7	ctrl_c_ctrl_sleep
8	dont_follow_me_pls
9	error_404_me_not_found
10	hash_me_if_you_can

11	idk_why_im_here
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14	maggi_in_2_hours
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16	not_a_robot_trust_me
17	procrastination_pro
18	sarcasm_loading
19	sleeping_is_my_cardio
20	too_lazy_to_think

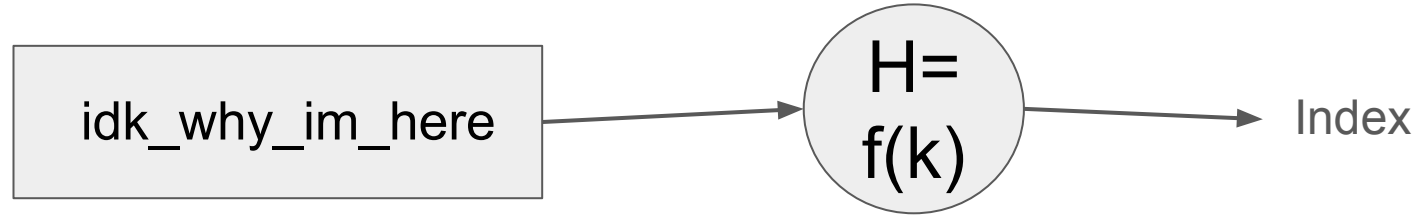
Hashing [Toy Example, say 10 users]



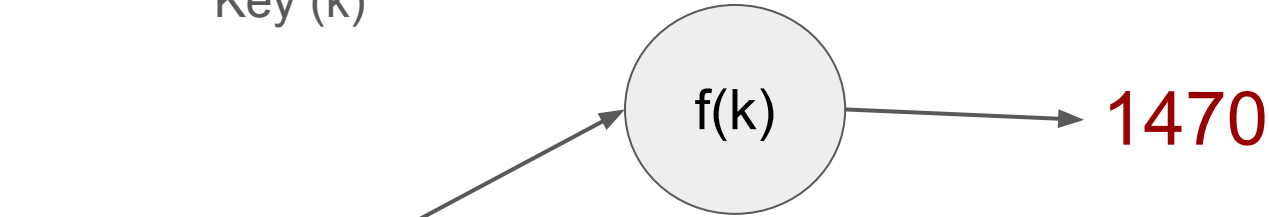
chai_over_wifi
sleeping_is_my_cardio



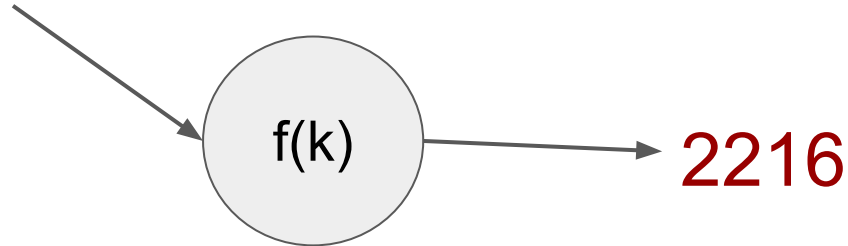
Hashing [Toy Example, say 10 users]



Key (k)



Chai_over_wifi
sleeping_is_my_cardio

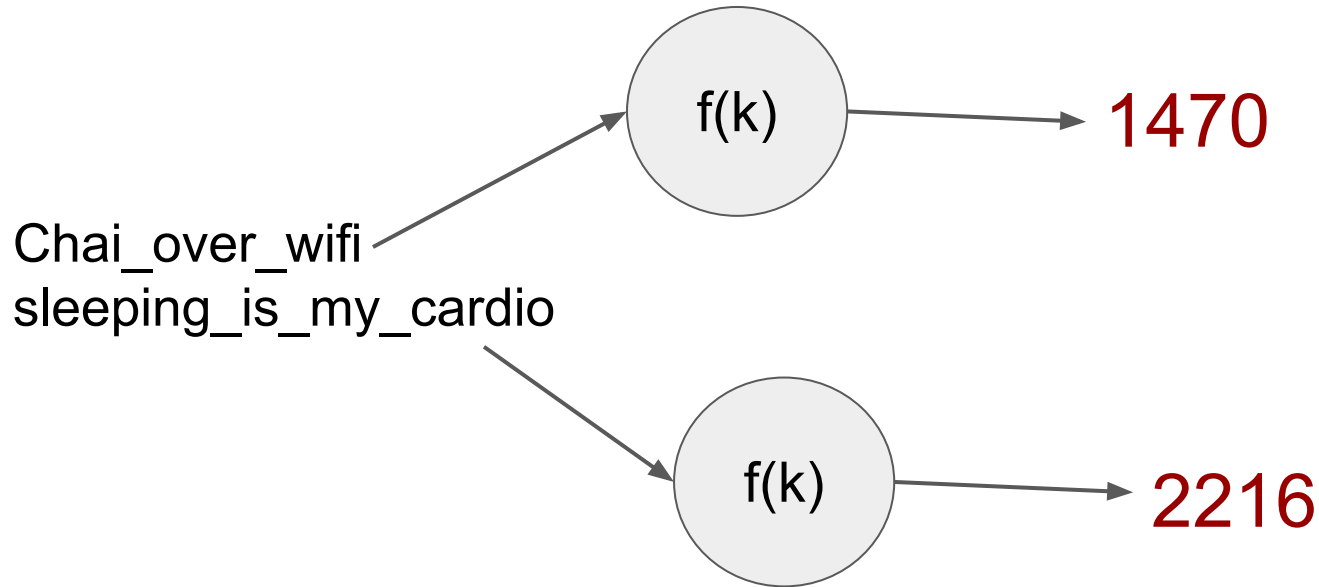


enough?

Hashing [Toy Example, say 10 users]

Let $S = \{a_1, a_2, \dots, a_m\}$ be a set of keys that must be stored in a table of size N .

Find a function such that, $H: S \rightarrow [0 \dots N - 1]$ that maps each key to a table index. Such a function is called a **hash function**.



Division Method [$H(k) = f(k) \bmod 10$]

0	idk_why_im_here	→ 1576
1	chai_over_wifi	→ 1470
2	maggi_in_2_hours	→ 1761
3	sleeping_is_my_cardio	→ 2216
4	error_404_me_not_found	→ 2156
5	born_to_nap	→ 1179
6	too_lazy_to_think	→ 1960
7	loading_username	→ 1705
8	crying_in_the_club	→ 1955
9	dont_follow_me_pls	→ 2035
10	sarcasm_loading	→ 1578

11	mood_swings_247	→ 1596
12	hash_me_if_you_can	→ 1934
13	bug_fixer_in_progress	→ 2191
14	ctrl_c_ctrl_sleep	→ 1933
15	caffeine_and_code	→ 1860
16	not_a_robot_trust_me	→ 2176
17	buffering_life	→ 1456
18	procrastination_pro	→ 2133
19	low_battery_human	→ 1949
20	brain_not_found	→ 1583

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“With just 20 usernames, we’re already seeing collisions. That’s not bad luck, that’s mathematics. Collisions are *far more likely than your intuition expects*.”

Birthday paradox: “In a room of just 23 people, there’s a 50% chance two share a birthday. Similarly, in hashing... you don’t need many keys before collisions become likely.”

ATM Transactions

Data Card Code (1) Discussion (0) Suggestions

Detail Compact Column

# totalBalance	# totalIncome	# totalOutco...
40000	0	0
40000	0	0
39490	670	1180
39760	575	815
40425	925	500
40150	640	490
40630	905	275
40145	745	600
40665	865	200
39235	0	765
39195	445	1250

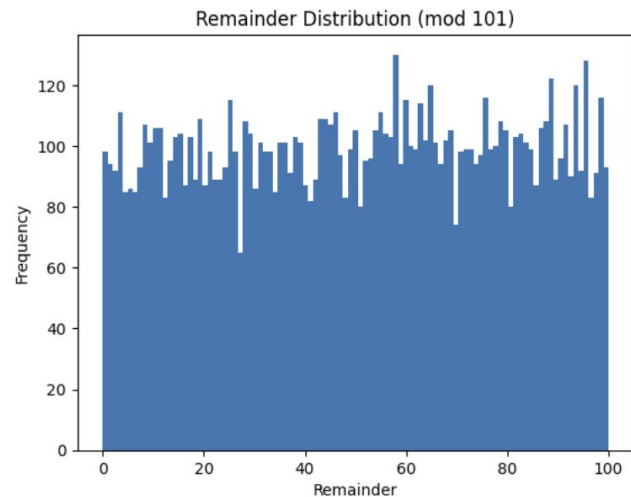
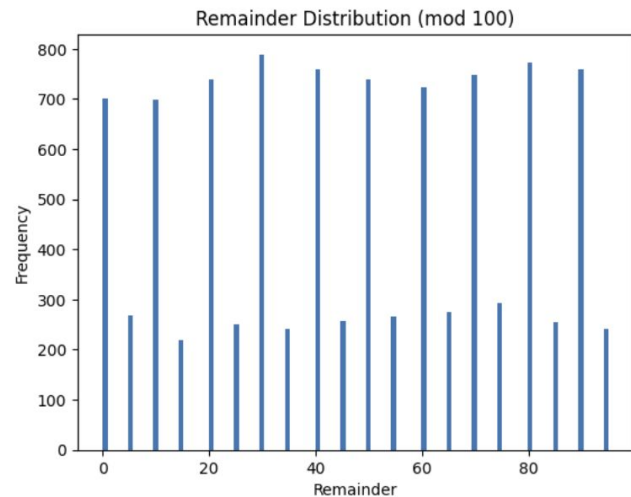
<https://www.kaggle.com/datasets/yamqwe/bulk-bookstore-dataset>

ATM Transactions

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# totalBalance	# totalIncome	# totalOutco...
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Popular Hashing Methods

Division Method

$$H(k) = k \bmod m$$

Consider a company each of whose 68 employees is assigned a unique 4-digit employee number. Suppose these needs to be stored at 100 two-digit addresses: 00, 01, 02, ..., 99. What will the address of each employ numbers 3205, 7148, and 2345. Assume $m = 97$.

Mid-Square Method

Consider a company each of whose 68 employees is assigned a unique 4-digit employee number. Suppose these needs to be stored at 100 two-digit addresses: 00, 01, 02, ..., 99. What will the address of each employ numbers 3205, 7148, and 2345. Assume 2 digits.

Folding Method

Consider a company each of whose 68 employees is assigned a unique 4-digit employee number. Suppose these needs to be stored at 100 two-digit addresses: 00, 01, 02, ..., 99. What will the address of each employ numbers 3205, 7148, and 2345. Assume fold of length 2.

Collision Resolution

Collision Resolution



Arjun Ananya
 Deepak

Aditi
Esha

Aarav
Harsh

Bhavya

Chetan

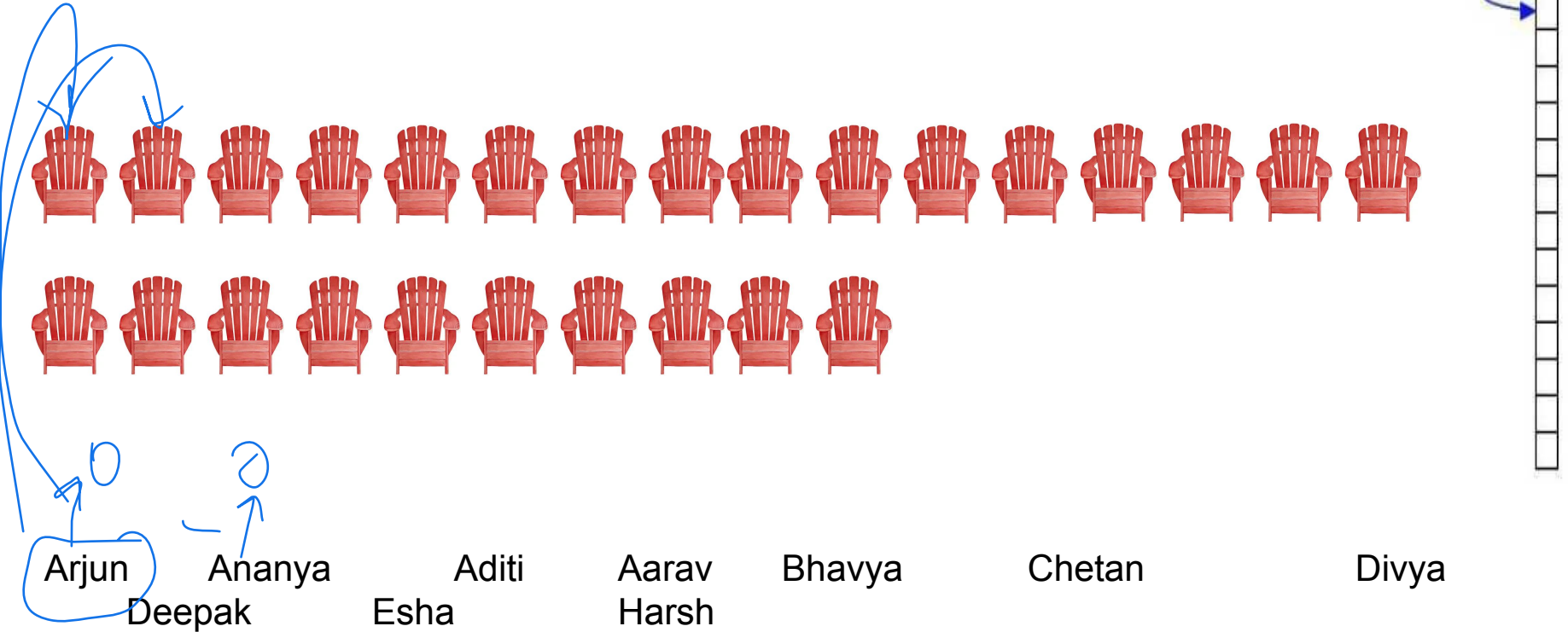
Divya

Collision Resolution with Chaining



Arjun	Ananya	Aditi	Aarav
	Bhavya	Chetan	Divya
Deepak	Esha	Harsh	

Linear Probing



Operation insert(x) : compute $K = f(x) \bmod N$, if the cell is full, find another by sequentially searching for the next available slot

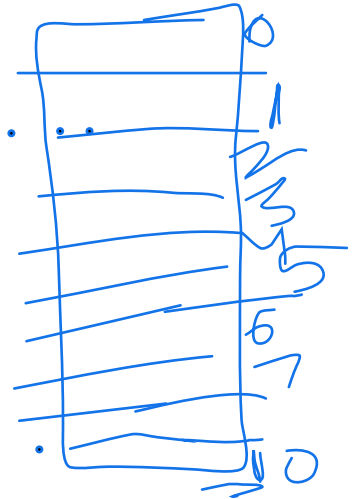
i.e. Go to $(K + 1) \bmod N$, $(K + 2) \bmod N$ etc.....

Operation find(x) : compute $f(x) \bmod N$, if the cell doesn't match, look elsewhere

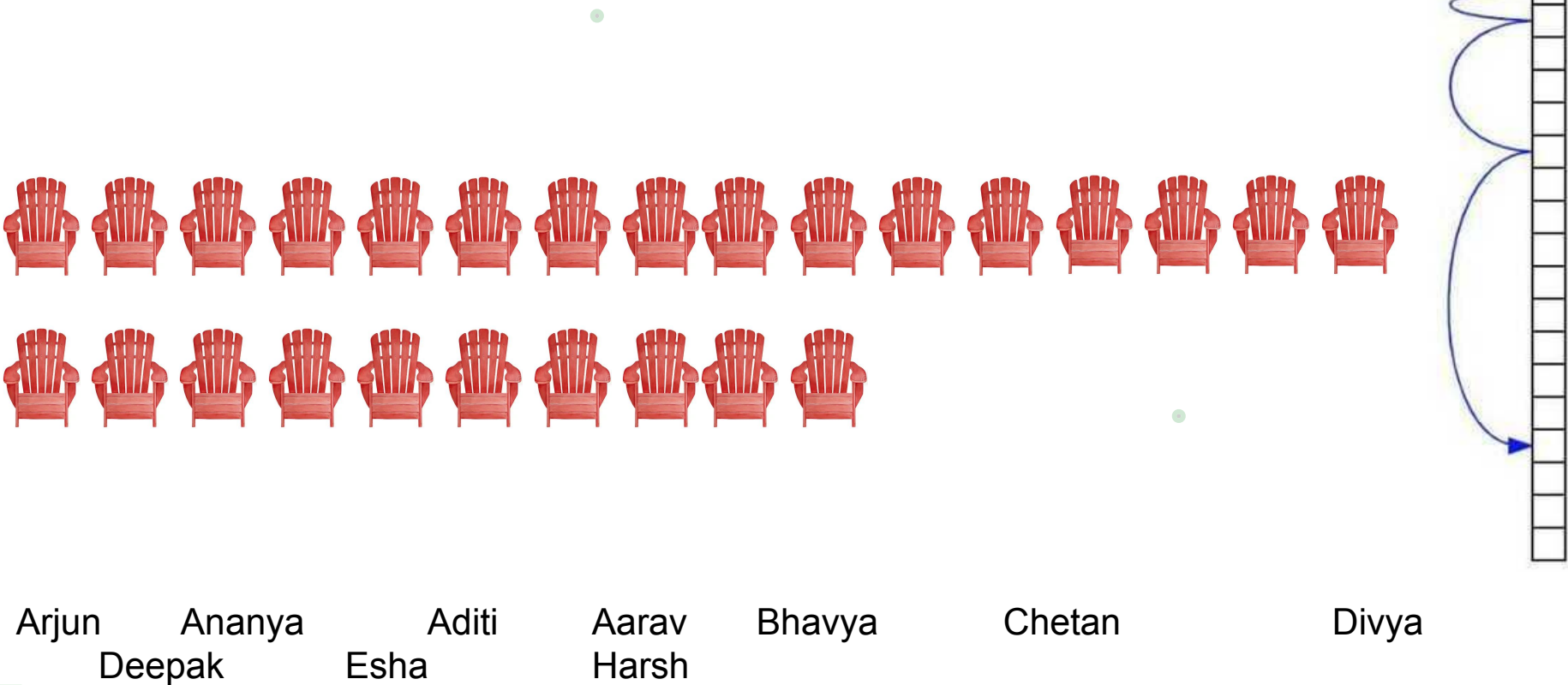
i.e. Linear probing function can be given by $h(x, i) = (f(x) + i) \bmod N$ ($i = 1, 2, \dots$)

54, 26, 93, 17, 77, 31, 44, 55, 20; $H(k) = k \bmod 11$

0	1	2	3	4	5	6	7	8	9	10
77	44	55	20	26	93	17			31	54



Quadratic Probing



Instead of checking the next slot linearly:

Define $h_0(k), h_1(k), h_2(k), h_3(k), \dots$

where $h_i(k) = (\text{hash}(k) + i^2) \bmod \text{size}$

$$(9 + 2^2) \bmod 11$$

54, 26, 93, 17, 77, 31, 44, 55, 20;

$$H(k) = k \bmod 11$$

0	1	2	3	4	5	6	7	8	9	10
77	44	20	55	26	93	17			31	54

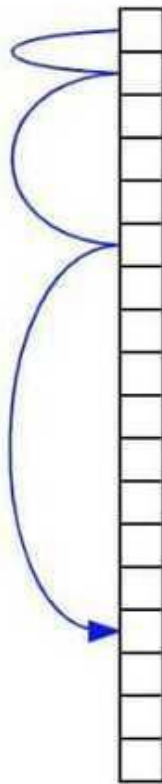
$$\begin{aligned} 0+0 \\ 0+1^2 \end{aligned}$$

$$\begin{aligned} 0 \\ 0+1^2 \\ 0+2^2 \end{aligned}$$

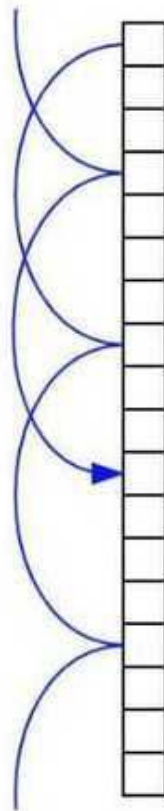
$$\begin{aligned} 0+3^2 \\ 0+4^2 \\ 0+5^2 \end{aligned}$$



Linear Probing



Quadratic Probing



Double Hashing

Suppose a record R with key k has the hash addresses $H(k) = h$ and $H'(k) = h'$

Probing Formula

$$h_i(k) = (H(k) + i \cdot H'(k)) \bmod m$$

$$(P + iS) \bmod (\text{Size})$$

54, 26, 93, 17, 77, 31, 44, 55, 20;

$$44 \bmod 11 = 0$$

$$H'(44) = 1 + 44 \bmod 10 = 5 \quad H(k) = 5$$

$$H(k) = k \bmod 11$$

$$H'(k) = 1 + (k \bmod 10)$$

$$54 \bmod 11 = 10$$

$$26 \bmod 11 = 4$$

$$93 \bmod 11 = 5$$

$$17 \bmod 11 = 6$$

$$77 \bmod 11 = 0$$

0	1	2	3	4	5	6	7	8	9	10
55	55	20	44	26	<u>93</u>	17			31	<u>54</u>

$$20 \bmod 11 = 9$$

$$31 \bmod 11 = 9$$

$$55 \bmod 11 = 0$$

$$H'(55) = 1 + 55 \bmod 10 = 6$$

$$H(20) = 1 + 20 \bmod 10 = 1$$